Adaptive Traffic Signal Optimisation

Simulation Experiment Report: Multi-Policy Evaluation vs BTP Fixed-Time Baseline

Simulator: SUMO v1.28.0 + TraCI | Baseline: BTP Marathahalli Bridge (210s) | Policies: Fixed, Webster, Adaptive, Pressure | Evaluated Seeds: 1–5 & Held-Out 11–15 | Scale: 1.7× Demand

This report is generated dynamically from simulation run results (`results/all_runs.csv`). It provides a rigorous, fair comparison across traffic demand scenarios between the fixed-time baseline and adaptive signal control policies.

1. Experimental Results & Scenario Comparison

1.1 Average Waiting Time per Vehicle (s)

Lower is better. Values represent Mean ± Standard Deviation over all evaluated seeds.

SCENARIO	FIXED (S)	ADAPTIVE (S)	ADAPTIVE VS FIXED (%)	PRESSURE (S)	PRESSURE VS FIXED (%)	FIXED_WEBSTER (S)	FIXED_WEBSTER VS FIXED (%)
balanced	43.03 ± 3.56	18.41 ± 4.71	-57.2%	15.62 ± 2.87	-63.7%	16.32 ± 3.46	-62.1%
imbalanced_ns	125.64 ± 4.60	44.47 ± 5.50	-64.6%	42.71 ± 3.62	-66.0%	40.18 ± 4.55	-68.0%
imbalanced_ew	151.78 ± 4.51	45.55 ± 5.73	-70.0%	43.86 ± 5.09	-71.1%	39.86 ± 3.14	-73.7%
surge_north	76.34 ± 8.31	55.22 ± 8.11	-27.7%	57.66 ± 5.77	-24.5%	54.83 ± 10.70	-28.2%

1.2 Average Queue Length (m)

Lower is better. Jam length measured back from the stop line by lane-area detectors.

SCENARIO	FIXED (M)	ADAPTIVE (M)	ADAPTIVE VS FIXED (%)	PRESSURE (M)	PRESSURE VS FIXED (%)	FIXED_WEBSTER (M)	FIXED_WEBSTER VS FIXED (%)
balanced	34.32 ± 4.12	15.16 ± 4.36	-55.8%	12.86 ± 2.65	-62.5%	14.59 ± 4.14	-57.5%
imbalanced_ns	60.50 ± 2.26	50.17 ± 3.12	-17.1%	49.12 ± 4.22	-18.8%	49.27 ± 3.29	-18.6%
imbalanced_ew	63.38 ± 1.36	48.11 ± 4.22	-24.1%	48.31 ± 6.61	-23.8%	49.02 ± 2.25	-22.7%
surge_north	60.32 ± 12.35	46.35 ± 4.53	-23.2%	48.49 ± 4.68	-19.6%	51.65 ± 11.47	-14.4%

1.3 System Throughput (veh/h)

Higher is better. Completed vehicles per hour during the evaluation window.

SCENARIO	FIXED (VEH/H)	ADAPTIVE (VEH/H)	ADAPTIVE VS FIXED (%)	PRESSURE (VEH/H)	PRESSURE VS FIXED (%)	FIXED_WEBSTER (VEH/H)	FIXED_WEBSTER VS FIXED (%)
balanced	2516.80 ± 29.38	2554.40 ± 20.85	+1.5%	2569.60 ± 33.06	+2.1%	2560.80 ± 27.00	+1.7%
imbalanced_ns	2392.00 ± 35.67	2777.60 ± 58.04	+16.1%	2789.60 ± 62.92	+16.6%	2736.00 ± 45.45	+14.4%
imbalanced_ew	2190.80 ± 28.59	2761.60 ± 25.27	+26.1%	2775.20 ± 35.74	+26.7%	2722.00 ± 32.62	+24.2%
surge_north	2785.60 ± 70.21	2822.40 ± 50.74	+1.3%	2815.60 ± 41.94	+1.1%	2778.80 ± 63.85	-0.2%

2. Performance Visualizations

Comparative performance charts generated dynamically across all demand scenarios and control policies:

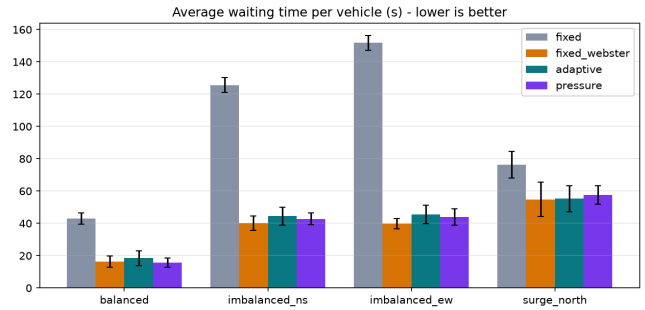


Figure 1: Average Waiting Time per Vehicle (s) by Scenario and Policy

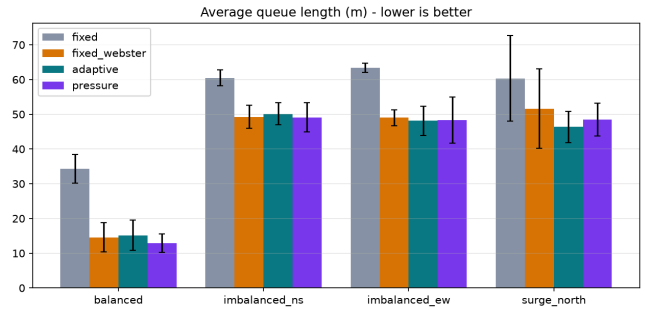


Figure 2: Average Queue Length (m) by Scenario and Policy

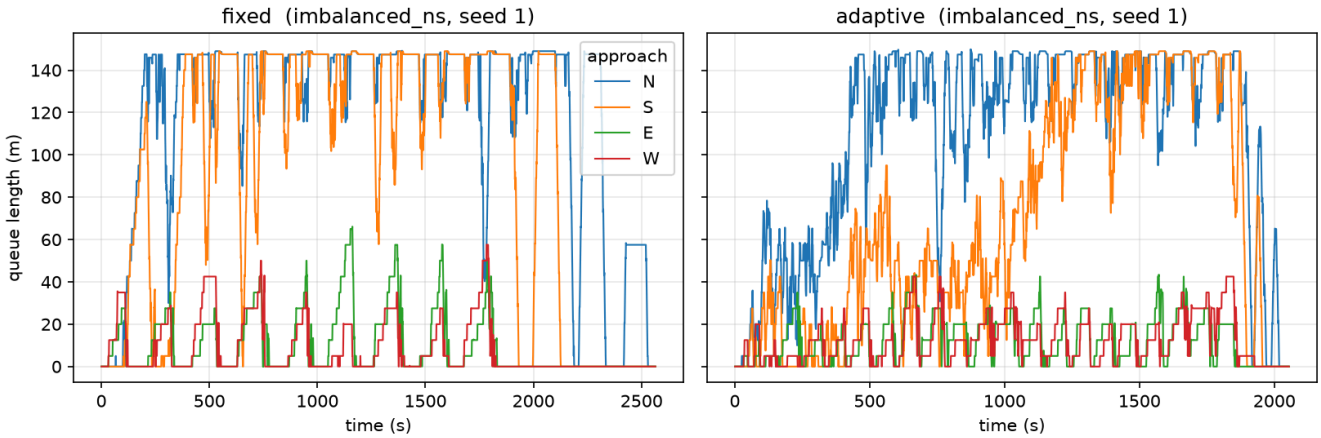


Figure 3: Queue Length Dynamics Timeseries Comparison across Approaches (Fixed vs Adaptive)

3. Known Limitations (docs/SPEC.md Section 9)

As explicitly documented in `docs/SPEC.md` section 9, the following limitations govern this phase of the study:

L1 **Approach Lane Geometry** Simulated 4-way cross junction with synthetic 2-lane approaches, without dedicated turn bays or real-world lane tapers.

L2 **Detector Sensing Cap** Lane-area detectors cover 150 m before the stop line; queues spilling beyond 150 m saturate sensor jam readings.

L3 **No Pedestrian Modelling** Pedestrian crossing phases are omitted (e.g., the 15 s exclusive pedestrian phase present in the BTP Marathalli plan is excluded from the modelled cycle).

L4 **Simple Two-Phase Plan** Traffic signal operates two alternating phases (NS and EW) with permissive turning rather than multi-phase protected right-turns.

L5 **Simulator vs Field Telemetry** All sensor measurements and driver behaviours originate from SUMO micro-simulation, not real-world street cameras or loop detector feeds.

4. Experimental Setup & Configuration Values

Evaluated Seeds: `[1, 2, 3, 4, 5]` | **Demand Scenarios:** `['balanced', 'imbalanced_ns', 'imbalanced_ew', 'surge_north']` | **Policies:** `['fixed', 'fixed_webster', 'adaptive', 'pressure']` | **Duration:** `1800 s` (Drain: `3600 s`) | **Scale:** `1.7`

Signal Timing & Clearance Invariants — `dp1.config.TIMING`

- Yellow Duration: `3 s`
- All-Red Clearance: `2 s`
- Safety Invariant: Strict `Green → Yellow → All-Red → Green`

Baseline Fixed-Time Controller — `dp1.config.FIXED`

- Calibration: BTP Marathahalli Bridge (Varthur x ORR)
- NS Green: `105 s` | EW Green: `95 s`
- Modelled Cycle: `210 s` (Phase 1: 110s, Phase 2: 100s)

Adaptive Queue Controller — `dp1.config.ADAPTIVE`

- Green Bounds: `10 s` min ≤ green ≤ `60 s` max
- Queue Scaling factor (`k_s_per_m`): `0.3 s/m`
- Gap-out Duration: `3 s`
- Extend / Competing Queue Thresholds: `15.0 m` / `1.0 m`

Adaptive Pressure Controller — `dp1.config.PRESSURE`

- Green Bounds: `10 s` min ≤ green ≤ `60 s` max
- Queue Difference factor (`k_s_per_m`): `0.3 s/m`
- Gap-out Duration: `3 s`
- Extend / Competing Queue Thresholds: `15.0 m` / `1.0 m`

Network Geometry & Webster Plan Configuration — `dp1.config.NET & WEBSTER`

- Lanes per approach: `2` (Lane 0: Right+Through, Lane 1: Left+Through)
- Approach / Detector length: `300.0 m` / `150.0 m`
- Speed limit: `13.89 m/s` (50 km/h)
- Webster saturation flow: `1800 veh/h/lane` (cycle clamped 40–120s)